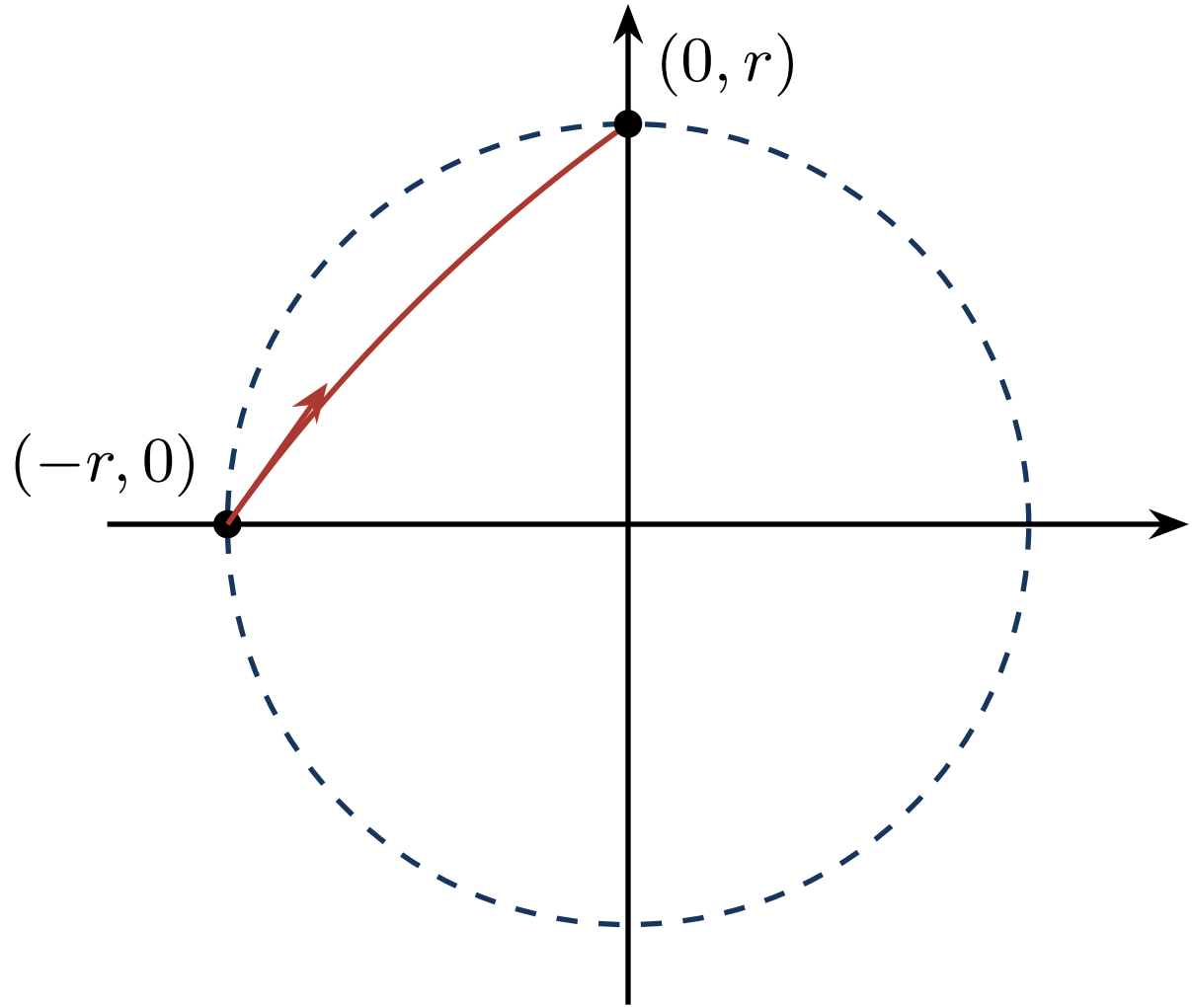

$$v_1 = \sqrt{3}\omega r\colon\ \alpha = 54.7356^\circ,\ \omega t_e = \pi/4$$